

PRIESTLEY FERNANDES

DATA & AI/ML AUTOMATION ENGINEER | NLP | GENAI | AGENTIC SYSTEMS

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SUMMARY

Results-oriented Data and AI/ML Automation Engineer with over 4 years of experience across financial services, healthcare, and consumer platforms. Skilled in building end-to-end AI systems using LLMs, agentic frameworks, NLP pipelines, and data automation tools. Adept at developing scalable ML models, automating claim audits, and deploying GenAI-driven applications for insight extraction, optimization, and productivity acceleration.

TECHNICAL SKILLS

- Programming & Tools: Python, R, SQL, SAS, Alteryx, Power Automate, Tableau, Power BI, Advanced Excel
- AI/ML & GenAI: Scikit-learn, PyTorch, TensorFlow, HuggingFace, LangChain, AutoGen, CrewAI, OpenAI APIs, Prompt Engineering
- Data & NLP Engineering: MongoDB, Oracle, Presidio, ETL Pipelines, LLM-RAG, spaCy, NLTK, Named Entity Recognition
- Statistical Analysis: Forecasting, Regression, Clustering, Factor Analysis, A/B Testing, ANOVA, Time Series Modeling

EXPERIENCE

Healthcare Retroactive Audits – Data and AI Automation Engineer | Jan 2025 – Present

- Automated claim auditing workflows using Alteryx, SQL, and OpenAI-based LLMs, cutting manual review time by 60% and enabling parallel document handling.
- Developed RAG-based resolution assistant using LlamaIndex and prompt templates, boosting processing precision by 40% across audit teams.
- Created secure PHI redaction system using spaCy, Presidio, and compliant LLMs, enabling compliant document pipelines for 100K+ records.
- Built LangChain + FAISS-powered Denials Bot to auto-generate insurance responses, increasing speed by 70% with structured JSON outputs.
- Engineered real-time EPIC Bot using concurrent workflows and Pydantic schemas to parse financial records and denial status in seconds.

Stable Coupons Inc – Data Scientist (AI/ML Engineering) | Feb 2024 – Dec 2024

- Built LLM-driven chatbot and recommender engine using scikit-learn and OpenAI, improving personalized results and increasing engagement by 40%.
- Integrated vector search via MongoDB Atlas and API gateway, improving semantic retrieval and backend throughput by 30%.
- Led analytics dashboard development with Power BI and Google Analytics to evaluate LLM-driven feature success.

Line Leverage – ML & Data Science Intern | Sep 2024 – Nov 2024

- Applied PCA and feature engineering to optimize NBA player datasets for model training and reduce noise by 20%.
- Built forecasting models (CNN-RNN, BiLSTM, XGBoost) using PyTorch and TensorFlow to support performance prediction and team analysis.

HeadOn – AI Intern | Aug 2024 – Nov 2024

- Developed retrieval systems with RAG architecture and OpenAI APIs to boost prompt relevance by 30% in politically-sensitive dialogue.
- Analyzed discourse sentiment and fairness using NLP techniques and trained classifiers to filter toxic and biased content.

Taipy.io – Data Analyst Intern | Jan 2023 – Aug 2023

- Cleaned and joined 60K+ logistics records for dashboarding using Pandas and Tableau, improving delivery insights by 50%.
- Built VRP optimization engine using Google OR Tools and Taipy UI, cutting cost by 18% through dynamic vehicle routing.

Infosys Limited – Systems Engineer | May 2018 – Feb 2021

- Upgraded FINACLE banking infrastructure for Union Bank of India across 9000+ branches, enhancing processing reliability by 12%.
- Developed and tested data pipelines using SQL, PL/SQL, and UNIX shell scripting to support custom trade finance modules.

EDUCATION

Stevens Institute of Technology – MSc in Business Intelligence and Analytics | Jan 2022 – Dec 2023

PROJECTS

- Finvest AI – LLM-powered personal finance agent developed during Lablab.ai hackathon using Llama3 and LangChain.
- PsyOzen AI – Mental health support bot using sentiment-aware prompt engineering and OpenAI's GPT models.
- ASL Detection – Built CNN-RNN pipeline to classify American Sign Language gestures in real-time video feeds.
- NVIDIA Time Series Forecasting – Used ARIMA and GARCH models to forecast volatility trends in NVIDIA stock.
- Reddit Anxiety Analysis – Mined subreddit networks to analyze mental health patterns using NLP and graph theory.
- California Groundwater Project – Conducted multivariate analysis of geospatial and climatic data to identify drought trends.
- Prototyped autonomous AI agents using LangChain, AutoGen, and CrewAI to route documents and generate action plans in real time.
- Benchmarked agent logic and feedback loops across workflows using simulated task chains and memory-enabled vector DBs.